

Nisan Harmanci

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EXPERIENCE

Quant Developer / Production Engineer

Oct 2025 – Present

Polymer Capital

Admiralty, Hong Kong

- Provide **first and second level support** for **20+ systematic trading strategies** across **6–7 PM** desks — covering equities and futures, handling real-time risk checks, order rejects, and execution monitoring.
- Sole engineering owner of two production **C#/Kafka** trading systems — leading migration to a high-performance **Redis/Postgres** execution framework on **AWS EC2**, processing real-time market data and risk checks with sub-second latency while maintaining zero downtime for live books.
- Use **Redis** for low-latency caching of risk and execution data across multiple strategies — reducing database lookups and accelerating order validation.
- Designed and engineered an emergency recovery protocol for recurring Kafka consumer lag issues generating **1000+ order rejects per book daily**. Isolated root cause as missed **FIX** messages during infrastructure outages; implemented an idempotent flush-and-reload script to clear backlogged pre-trade risk constraints, restoring full order flow within **20 minutes** and eliminating market impact from expired orders.
- Architected and deployed a **Python**-based production alerting system that consolidates fragmented Grafana alerts into a unified real-time Teams **Workflow** — fetching all risk/order metrics per 5-minute cycle, then routes data to specialized checkers for GMV breaches, order volume anomalies, and notional rejects — eliminating redundant DB calls and enabling low-latency alert propagation.
- Reduced incident detection time from **30+ minutes** to **sub-5-minute** resolution by replacing stale terminal-based alerts and midnight false-positives with the custom alerting framework — now the sole monitoring solution for the new trading system and used desk-wide.
- Built and maintain internal dashboards in **Grafana and Streamlit** that allow the production team to visually monitor system health and manage risk parameters directly via UI controls — eliminating manual server access and reducing operational overhead.
- Own and operate the **Apache Airflow** orchestration layer for two critical market data sources — managing daily commission targets calculation and trading hours evaluation to support systematic trading operations.
- Onboard new strategies into production — working with Middle Office, Compliance, and external vendors to set up risk limits, clear regulatory checks, and ensure smooth production integration.
- Manage **FIX** message ingestion from external OEMS (TORA) into production systems, automating order validation and risk checks for PM batch scripts and algorithmic strategies.
- Spearheaded end-to-end strategy deployment on **AWS EC2** — from provisioning servers and configuring infrastructure to integration with pre-trade risk, alerting, and order execution management systems.
- Support deployment, checkout, and **QA testing** of trading applications — integrating with the data/infra team (Kubernetes-managed PM pods, position inventory managed centrally) to ensure smooth production releases.

Project Research Assistant

Jul 2023 – May 2024

LSCM R&D Centre

Cyberport, Hong Kong

- Developed **Python** web scrapers extracting structured data from 10,000+ webpages to seed the project's first dataset.
- Engineered efficient data analysis pipelines for over 40,000 JSON-formatted textual tokens.
- Boosted model accuracy by **38%** by selectively implementing dimensionality reduction and clustering algorithms.
- Optimised database search performance by **100%** by deploying **Milvus** on **Kubernetes**.

SKILLS & CERTIFICATIONS

Languages: Python, C#, C++, SQL, Bash/Shell

Tools & Infra: Linux, Windows, Git, Docker, Kubernetes, Ansible, AWS (EC2, S3), Apache Airflow, Grafana, Streamlit, Kafka, Redis, Postgres

Domain: FIX Protocol, Order Management Systems (OMS), Pre-Trade Risk, Execution Management, Incident Management, Production Support, Full SDLC, System Design, Distributed Systems, API Development, Post-Trade Technology, Real-Time Systems

EDUCATION

City University of Hong Kong

B.Sc. in Computer Science

Hong Kong

2021 – 2025